

Yanming Zhang

yanmzhang@cs.stonybrook.edu | [LinkedIn](#) | [Google Scholar](#)

Trustworthy AI, Machine Learning, Causal Inference, Visual Analytics

SKILLS SUMMARY

- **Programming Languages:** C++, Java, Python, JavaScript, R, GO, MATLAB
- **Tools:** PyTorch, TensorFlow, Numpy, Pandas, Scikit-learn, MySQL, Git, Docker, D3.js, React, Flask
- **CS Fundamentals:** Machine Learning, Deep Learning, Large Language Model, Mathematical Modeling

EDUCATION

- **Stony Brook University** Stony Brook, New York
Ph.D. in Computer Science; GPA: 3.90 *Aug 2020 - Present*
Courses: Machine Learning, Computer Vision, Probability and Statistics for Data Science, Data Science, Network Security

EXPERIENCE

- **Netflix** Los Gatos, California
ML/AI Intern *Jun 2026 - Present*
 - **Algorithm Explainability Research:** Built an observability and explainability framework for Netflix's homepage row-level recommender, improving visibility into ranking behavior and performance.
- **iMouse GmbH** Remote
Summer Research Intern *Jun 2025 - Aug 2025*
 - **Vision Language Model Research:** Built and integrated multimodal models for video-to-text summarization of abnormal mouse behavior. Leveraged multimodal RAG and LoRA fine-tuning of VLMs, with PoC implementations.
- **Nokia Bell Labs** Murray Hill, New Jersey
Summer Research Intern *Jun 2019 - Aug 2019*
 - **Computational Imaging Research:** Collaborated with researchers to analyze light geometry and signal behavior of a volumetric imaging device, and developed a 3D imaging simulation framework using MATLAB and Python.

SELECTED PROJECTS

- **Causal Analysis of Aha Moments in LLM Reasoning** *Apr 2026 - Present*
 - **Causal Explanation of Aha Moments:** Extended causal explanation techniques from sequential decision-making to LLM reasoning, constructing causal models of latent states, uncertainty, reasoning transitions, and task outcomes to characterize mechanisms underlying Aha moments.
 - **Intervention for Efficient Reasoning:** Investigating counterfactual and interventional strategies to identify causal precursors of beneficial reasoning transitions and induce earlier self-correction, with the goal of improving reasoning accuracy while reducing unnecessary inference steps.
- **Large Language Models in Visual Analytics** *Jun 2023 - Present*
 - **LLM-Based Causal Hypothesis Evaluation:** Engineered LLM-powered methods to evaluate causal hypotheses and identify latent variables, enhancing user comprehension through integrated visualizations and natural-language justifications; validated performance through ablation studies.
 - **Persona-based Causal Model Exploration:** Modeled and evaluated LLM personas to analyze how persona design and contextual themes influence causal model construction and downstream behavior.
- **Explainable Reinforcement Learning in Real Strategy Games** *Jun 2022 - Sep 2024*
 - **Multi-Level Agent Explanations:** Created post-hoc explanation methods, including counterfactual trajectories for tactical insights and high-level causal models for strategic insights, enabling explanation generation of agents' decisions.
 - **XRL Dashboard Development:** Developed an explainable reinforcement learning dashboard that leverages visual analytical techniques to improve human understanding of RL agents' decision-making processes.

SELECTED PAPERS

- **CausalXRL: Explainable Reinforcement Learning through Causal Graph Reasoning:**
Y. Zhang, E. Papenhausen, K. Mueller in ICML 2026 (Jul' 2026)
Proposed a novel methodology for explainable reinforcement learning that generates context-aware causal explanations for model-free RL agents using human-understandable bottlenecks, improving performance and examined across 4 RL benchmarks.
- **Fairness Beyond Populations: Subgroup-Specific Causal Diagnosis for Algorithmic Decision Auditing:**
Y. Zhang, K. Mueller in ACM EAAMO 2026(Jul' 2026)
Presented an interpretable workflow for discovering and explaining subgroup-level disparities through a causal lens, revealing patterns that fairness metrics alone cannot detect.
- **CausalChat: Interactive Causal Model Development and Refinement Using Large Language Models:**
Y. Zhang, A. Kota, E. Papenhausen, K. Mueller in IEEE Transactions on Visualization and Computer Graphics (Aug' 2025)
Developed a full-fledged pipeline for LLM-assisted causal refinement, realized through an interactive visualization interface, and validated its effectiveness via experiments engaging both expert and non-expert users.

SELECTED PRESENTATIONS

- **Disentangling Legal and Illegal Opioid Risk Factors Over Time:A County-Level Study** Nov 2025
American Public Health Association 2025 Annual Meeting and Expo *Washington, DC*